

Chapter 12

Problems 3

In this, short, chapter we will solve only one problem, a problem that deals with the so-called Chain Of Pappus. However, the length of this chapter is deceiving because just this one problem alone punches way above its weight. In order to see that, we highly recommend that our readers try solve this problem in any which other way, a way that does not rely on the machinery of inversive geometry.

Being realistic and with all due respect, the chances are high that most of our readers will fail do so and it is that, good and educational, failure is what brings to light the awesome power of the “Transform And Conquer” problem-solving approach in general and the inversive geometry in particular.

Transform. Solve. Reverse-transform.

Problem 18 (the Chain Of Pappus). Given two distinct coplanar circles, $q_1(Q_1, R_1)$ and $q_2(Q_2, R_2)$ ($R_1 > R_2$), that touch each other internally at a single point, T , let an infinitude of smaller circles be inscribed in the space between the circles q_1 and q_2 in such a way that each inscribed circle touches two of its immediate neighbors and the given smaller circle, q_2 , externally, while touching the given larger circle, q_1 , internally.

Challenges:

- express the distance from the center of the n -th inscribed circle to the straight line, l , through the centers of the given circles q_1 and q_2 in terms of the length of its radius r_n and n
- express the length of the radius r_n in terms of the lengths of the radii R_1 , R_2 and n

Solution. At first blush, this problem seems to be either impenetrable or daunting.

As far as we are aware, [Pappus of Alexandria](#) actually did manage to find an elementary path to victory in this case, but the said solution found by Pappus is not that easy on the eye.

In contrast, the employment of a wisely chosen inversion of the parent plane with respect to a circle leads to almost a kindergarten variety of a deduction.

Indeed.

Let us choose the single point of tangency of the two given circles, T , as the center of inversion.

Let us choose the magnitude of $2R_1$ as the radius of inversion.

Moreover, let us analyze where the relevant objects of this problem will be taken under the inversion of the parent plane with respect to the circle $c(T, 2R_1)$ with positive power.

Well, under the said inversion:

- due to Theorem 9, the straight line l that passes through the centers of the given circles, Q_1 and Q_2 , will be taken into itself
- due to Theorem 17, the given circle q_1 will be taken into a straight line, q'_1 , that will be tangent to the circle of inversion, c
- due to Theorem 19, the given circle q_2 will be taken into a straight line, q'_2 , that will have no points in common with the circle of inversion, c
- because both given circles, q_1 and q_2 , are orthogonal to the straight line l , both straight lines q'_1 and q'_2 will also be orthogonal to the straight line l , which, in turn, means that in a Euclidean plane the said straight lines, q'_1 and q'_2 , will be parallel
- therefore, the family of the original inscribed circles, k_n , will be taken into a family of the inverted circles, k'_n , all of which, due to the preservation of the tangency points under an inversion of the real plane with respect to a circle, will remain tangent to the parallel straight lines q'_1 and q'_2

From the last observation above it follows that the lengths of all the radii r'_n of the inverted circles k'_n will be the same. Hence, we might just as well drop the sub index n in the designation of the length of the radius of the n -th inverted circle r'_n and refer to such a length simply as r' .

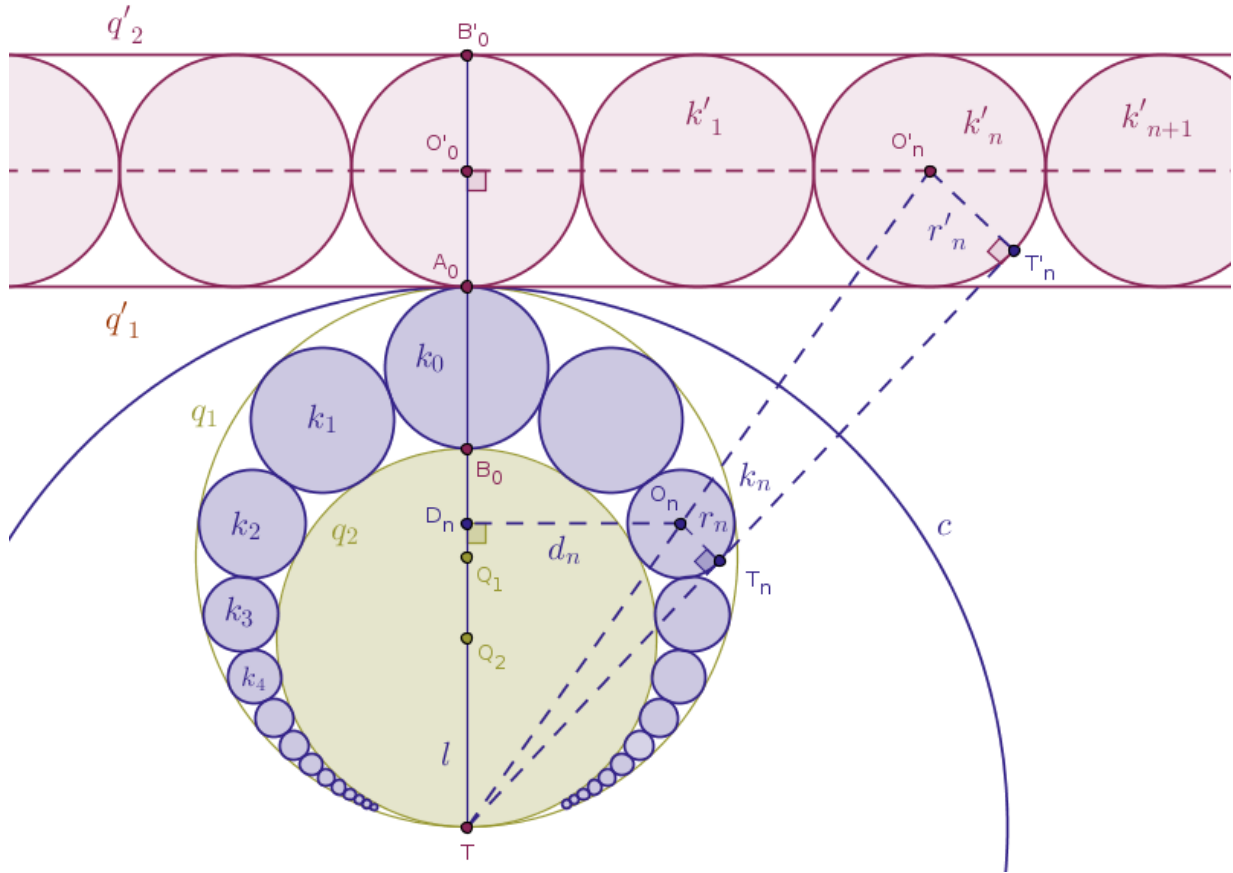
Thus, for the family of the inverted circles k'_n the expression that ties all the inversive equivalents of the magnitudes sought-after is less than trivial:

$$|O'_0 O'_n| = d'_n = n \cdot 2 \cdot r'_n$$

from where:

$$\frac{d'_n}{r'_n} = 2n \quad (1)$$

In the last two relations we took it that n is a natural number that is used to mark the ordinal number of the original circle with respect to the line of centers l . The original circle that we will refer to as k_0 with $n = 0$ has its center positioned **on** the said straight line l , the immediate right or left neighbor of the circle k_0 will be the original circle k_1 with $n = 1$ and so on (Figure 12.1):



The corresponding inverted circles k'_n will be numbered in the same way via k'_0, k'_1, k'_2 and so on. We show the relevant sub scripts here for clarification and demonstration purposes. As we already agreed, we will drop the said index in the consequent manipulations.

As such, using the relation (1) as our workhorse, we would like to express the magnitudes of d'_n and r'_n in terms of d_n and r_n . Thanks to the plethora of right triangles, that task is of middle school variety.

Consider the triangles TD_nO_n and $TO'_nO'_n$ which are not only right by construction but also share a common angle at the vertex T . By **AAA**, these triangles are similar and, hence, it is the case that the lengths of their corresponding sides are in the same proportion:

$$\frac{d_n}{|TO_n|} = \frac{d'_n}{|TO'_n|}$$

which is to say that:

$$\frac{|TO_n|}{|TO'_n|} = \frac{d_n}{d'_n}$$

Under the inversion with respect to our circle $c(T, 2R_1)$ with positive power a straight line passing through the point T that is tangent to the circle k_n at the point T_n will be transformed into itself, while the point T_n will be carried into its inverse point T'_n . Since the tangency points under any inversion of the real plane with respect to a circle are preserved, it follows that the straight line TT_n will also touch the inverted circle k'_n at the point T'_n .

Hence, the right triangles TO_nT_n and $TO'_nT'_n$, which share a common interior angle at the vertex T , are similar by **AAA** and the lengths of their corresponding sides are also in the same proportion:

$$\frac{|TO_n|}{r_n} = \frac{|TO'_n|}{r'}$$

which is to say that:

$$\frac{|TO_n|}{|TO'_n|} = \frac{r_n}{r'}$$

from where it follows that:

$$\frac{r_n}{r'} = \frac{d_n}{d'_n}$$

On the other hand, from (1) we have:

$$\frac{r_n}{d_n} = \frac{r'}{d'_n} = \frac{1}{2n}$$

from where we get a surprisingly succinct expression for the sought-after distance d_n of the center of the n -th inscribed circle from the line of centers Q_1Q_2 in terms of the length of the radius of the said inscribed circle r_n :

$$\boxed{d_n = 2nr_n} \quad (2)$$

Before we express the length r_n of the n -th inscribed circle in terms of the lengths of the radii R_1, R_2 and n , let us compute the distance $l_n = |TD_n|$, which is the distance from the tangency point T to the projection D_n of the center O_n of the n -th inscribed circle on the line of centers l passing through the points Q_1 and Q_2 .

From the similar right triangles TD_nO_n and $TO'_0O'_n$ it follows that:

$$\frac{l_n}{d_n} = \frac{|TO'_0|}{|O'_0O'_n|} = \frac{|TO'_0|}{2nr'}$$

from where:

$$l_n = \frac{d_n}{2n} \cdot \frac{|TO'_0|}{r'}$$

From (2) we have:

$$\frac{d_n}{2n} = r_n$$

and hence:

$$l_n = r_n \cdot \frac{|TO'_0|}{r'} \quad (3)$$

Next, we need to express the lengths $|TO'_0|$ and r' via the lengths of the given radii R_1 and R_2 . To that end, the length of the line segment TO'_0 can be represented as a compound quantity:

$$\begin{aligned} |TO'_0| &= 2R_1 + \frac{|A_0B'_0|}{2} = \\ &= 2R_1 + \frac{|TB'_0| - 2R_1}{2} \end{aligned}$$

which is to say that:

$$|TO'_0| = R_1 + \frac{|TB'_0|}{2} \quad (4)$$

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But the point B'_0 is the inverse image of the point B_0 with respect to the circle $c(T, 2R_1)$ with positive power. Hence, due to Condition 3 of Definition 1, for these two points we have:

$$|TB'_0| \cdot |TB_0| = |TB'_0| \cdot 2R_2 = 4R_1^2$$

meaning that:

$$|TB'_0| = \frac{2R_1^2}{R_2} \quad (5)$$

Putting the RHS of (5) back into the RHS of (4), we find:

$$|TO'_0| = R_1 + \frac{2R_1^2}{2R_2}$$

or:

$$|TO'_0| = \frac{R_1 \cdot (R_1 + R_2)}{R_2} \quad (6)$$

The length of the radius r' of the family of the inverted circles k'_n is, of course, just one half of the distance between the points A_0 and B'_0 :

$$r' = \frac{|A_0B'_0|}{2} = \frac{|TB'_0| - 2R_1}{2}$$

which, with (5), means that:

$$\boxed{r' = \frac{R_1 \cdot (R_1 - R_2)}{R_2}} \quad (7)$$

Putting the RHS of (7) back into the RHS of (3), we find the length l_n in terms of the lengths of the radii R_1 , R_2 and r_n :

$$\boxed{l_n = r_n \cdot \frac{R_1 + R_2}{R_1 - R_2}} \quad (8)$$

Note that the relation (8) meets the first challenge of the problem posed.

In order to meet the second challenge of the problem posed, let us now express the length of the radius r_n in terms of the lengths of the radii R_1 , R_2 and n as requested.

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From the similarity of the right triangles TO_nT_n and $TO'_nT'_n$ it follows that:

$$\frac{r_n}{|TT_n|} = \frac{r'}{|TT'_n|}$$

implying that:

$$r_n = r' \cdot \frac{|TT_n|}{|TT'_n|} \quad (9)$$

Applying the Condition 3 of Definition 1 to the inverse images of each other T_n and T'_n , we obtain:

$$|TT_n| \cdot |TT'_n| = 4R_1^2$$

which is to say that:

$$|TT_n| = \frac{4R_1^2}{|TT'_n|}$$

and hence:

$$r_n = r' \frac{4R_1^2}{|TT'_n|^2} \quad (10)$$

where the magnitude of r' in the suitable shape is already known to us from (7).

In order to compute the magnitude of the square $|TT'_n|^2$, we recall that in this particular case that magnitude represents none other than the power of the point T with respect to the inscribed circle k'_n :

$$|TT'_n|^2 = |TO'_n|^2 - r'^2 \quad (11)$$

where the magnitude of the square $|TO'_n|^2$ can be obtained via the Pythagorean theorem applied to the right triangle $TO'_0O'_n$:

$$|TO'_n|^2 = |TO'_0|^2 + 4n^2r'^2$$

But we already found the length of the line segment TO'_0 while computing the length l_n , see the relation (6):

$$|TO'_0|^2 = \frac{R_1^2 \cdot (R_1 + R_2)^2}{R_2^2}$$

Thus, using the expression for r' from (7), we obtain:

$$|TO'_n|^2 = \frac{R_1^2 \cdot (R_1 + R_2)^2}{R_2^2} + 4n^2 \cdot \frac{R_1^2 \cdot (R_1 - R_2)^2}{R_2^2} \quad (12)$$

Now that we have recovered the magnitude of the square $|TO'_n|^2$, we put the RHS of (12) back into the RHS of (11) in order to recover the magnitude of the square $|TT'_n|^2$:

$$|TT'_n|^2 = \frac{R_1^2 \cdot (R_1 + R_2)^2}{R_2^2} + 4n^2 \cdot \frac{R_1^2 \cdot (R_1 - R_2)^2}{R_2^2} - \frac{R_1^2 \cdot (R_1 - R_2)^2}{R_2^2}$$

which, after factoring out the square of the R_1/R_2 ratio, comes out to be:

$$|TT'_n|^2 = \frac{R_1^2}{R_2^2} \cdot \left((R_1 + R_2)^2 + (4n^2 - 1) (R_1 - R_2)^2 \right) \quad (13)$$

Putting the magnitudes of r' from (7) and $|TT'_n|^2$ from (13) back into (10), we obtain:

$$r_n = \frac{R_1 \cdot (R_1 - R_2)}{R_2} \cdot \frac{R_2^2}{R_1^2} \cdot \frac{4R_1^2}{(R_1 + R_2)^2 + (4n^2 - 1) (R_1 - R_2)^2}$$

Canceling the single copy of the product $R_1 \cdot R_2$ in both the numerator and the denominator of the above fraction, we arrive at the expression of the length r_n of the radius of the n -th inscribed circle in terms of the given radii R_1, R_2 and n sought-after:

$$r_n = \frac{4R_1 \cdot R_2 \cdot (R_1 - R_2)}{(R_1 + R_2)^2 + (4n^2 - 1) (R_1 - R_2)^2} \quad (14)$$

Finally, we can recover the lengths l_n and d_n in terms of R_1, R_2 and n .

Putting the RHS of (14) back into the RHS of (8), we recover the length l_n :

$$l_n = \frac{4R_1 \cdot R_2 \cdot (R_1 + R_2)}{(R_1 + R_2)^2 + (4n^2 - 1) (R_1 - R_2)^2} \quad (15)$$

Putting the RHS of (14) back into the RHS of (2), we recover the length d_n :

$$d_n = \frac{8nR_1 \cdot R_2 \cdot (R_1 - R_2)}{(R_1 + R_2)^2 + (4n^2 - 1) (R_1 - R_2)^2} \quad (16)$$

Observe that during the solution of this particular problem we did engage in some symbolic shuffling but the intellectual voltage of the said shuffling was hovering right around zero because it was just a finite computation.

All the heavy conceptual lifting was done by the idea of applying the inversive geometry to a particular setting.

Also, because we did not want to break up the flow of the solution, we did not mention it then, but we shall mention the following fact now: our early bulleted analysis of which original object is taken into which image under the chosen inversion is a typical step during the solution of a problem via the application of inversive geometry to it.

Oftentimes, we would engage into a sort of rapid-fire analysis of what goes where under this inversion with the sole purpose of trying to get a gut feeling about whether this inversion is promising or not. If this inversion does not look promising or fruitful then we quickly switch to that inversion and repeat the process all over again.

Because a universal problem-solving algorithm does not exist, such a try-fail hopping around, of course, does not guarantee a success, only sometimes that process may lead to victory. $\square$